nature portfolio

Peer Review File

Self-oscillating synchronematic colloids

Corresponding Author: Professor Kyle Bishop

This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)
Through simulations and experiments, the authors study Quincke colloids, which behave as self-oscillating units (active oscillators). In this study, different collective states are observed, including temporal synchronization and directional alignment, termed "synchronematic order". The authors distinguish between fluid-like clusters, where this order is short-range, and crystalline clusters, where the system is characterized by global synchronization and circular alignment.

This paper reveals interesting collective phenomena arising in Quincke colloids. The manuscript is well-written, even if a careful revision of all the formulas (mainly in the methods) is needed -- several typos in the equations are present. This paper is potentially suitable for publication in Nature Communications only if the authors address my questions and clarify the mechanisms generating their experimental results.

- I believe that this paper would benefit from a restyling: the authors should introduce sections, distinguishing between introduction, results (perhaps with subsections), and discussion.

- Could the authors clarify what they mean when they write "these spatiotemporal structures arise from different initial conditions ..." or "By varying the initial particle distribution ..."? This point should be explained with more details in the main text and methods. Do these different phases emerge when different packing fractions are considered? The authors should clarify the experimental limitations: are the authors able to control the density in their system? What do the authors mean by the sentence "when initialized within local regions of high density ..."?

- The fluid-cluster snapshots (Fig.1 e and Fig.2 a) do not show a cluster with a fluid structure but (apparently) a homogeneous fluid. I suggest that the authors show both an image where the cluster is evident and, additionally, the already present zoom, which focuses on the internal structure (Fig. 2 a).

- The phase-angle spatial order (exponential decay) resembles the short-range order observed in the velocity field of self-propelled systems (without alignment interactions):
* L Caprini, U Marini Bettolo Marconi, A Puglisi. Phys. Rev. Lett. 124 (7), 078001, 2020 "Spontaneous velocity alignment in motility-induced phase separation"
Could the authors comment on these previous results and provide an intuitive argument to explain the origin of the observed phenomenon?

- Concerning the synchronematic crystalline phase, is the phenomenon observed a result of finite-size effects? In other words, does this effect disappear for larger cluster sizes? Alternatively, one can wonder if, in the absence of cluster boundaries -- say, by considering a homogeneous crystal (in simulations) with periodic boundary conditions -- the same synchronization effect appears. I expect that this will not be the case: this point should be clarified. Somehow, this state is reminiscent of the self-reverting vorticity observed in clusters consisting of chiral active particles:
* L Caprini, B Liebchen, H Löwen. Commun. Phys. 7 (1), 153, 2024 "Self-reverting vortices in chiral active matter"
The authors may comment on this analogy in the final discussion.

Minor points:
- Along with the paper, the authors mention "close-packed oscillating crystals". Actually, I believe that it is more correct that the authors refer to this state as "oscillating clusters with a crystalline structure".

Typos:
- On page 5, some formulas in line with the text are corrupted, as the formula for r and theta.
- Typo in the Methods: the section title is "Data nalysis" rather than "Data Analysis".
- Several typos are present in Eqs.(2),(3),(4), and (5).
- In formula (8), the minus sign in the argument of the exponential is missing, as well as the minus in the delta Dirac function after Eq.(7).

(Remarks on code availability)

Reviewer #2

(Remarks to the Author)
The present manuscript is a combined experimental/theoretical study concerning the collective dynamics of oscillatory Quincke rollers. Specifically, the authors study spherical particles that, under an applied electric field, exhibit a back-and-forth rolling motion characterized by frequency, phase, amplitude, and axis n. They find that a semi-dilute monolayer of these particles can form liquid-like phases with short-ranged synchronization and clustering (Fig. 1e, Fig. 2b). A more dense monolayer forms rocking crystallites with circular alignment of the particles' rocking directions (Fig. 3b). Numerical modeling reveals that the behavior arises from the interplay of hydrodynamic interactions and dipole-dipole interactions, with an additional role for electroosmotic attraction for the crystallites. Excellent quantitative agreement (Fig. 3d-e) is obtained between the numerics and experiment with seemingly limited fitting.

The study seems thorough and scientifically sound (although I have some questions, as detailed below). Turning to the question of impact/significance: while the actual rocking motion seems, to honest, underwhelming in videos, what I judge the authors to have achieved in this manuscript is linking temporal (synchronization) and spatial (clustering, alignment) forms of ordering. I think this will open new directions in out-of-equilibrium colloidal self-organization. However, to make this case more effectively, the authors need to situate their work with respect to previous work on "swarmalators" and chiral active matter: the manuscript is missing, e.g., citations of D. Levis and J. Lintuvuori. How does the present work go beyond previous work that links flocking and synchronization in circle swimmers? (In addition, I suggest citing the work on AC Quincke rollers from Sano et al.)

As a second point: I find the role for hydrodynamic interactions to be very intriguing, but obscured by the presentation of the manuscript. HI are discussed either superficially in the main text (e.g., a cursory mention of vorticity) or buried in formalism/methodology. Overall, a mechanistic understanding of the pairwise interaction of two rollers (interplay of dipole/dipole interactions and HI) is missing. While dipole/dipole interactions are at least well understood, I suggest that the authors could provide at least provide data on the (time-averaged?) flow and vorticity fields sourced by a roller. How do HIs in this instance differ from, e.g., the usual pusher/puller stresslet HIs?

More detailed comments/questions:
• Section headings might help with navigating this manuscript.
• Please elaborate on the point on p. 5 about the value of the covariance suggesting a common origin of synchronization and alignment.
• Please clarify the argument on p. 9 concerning why the finite amplitude of oscillations limits the diameter for rigid body motion. (Where does the factor of \pi come from?)
• Concerning the role for "q" in the numerics: is it correct that q \neq 0 in Eqs. 2-4, since there is represents surface charge, but is assumed to be zero for particle/particle interaction, due to screening by ions?
• The introduction of Eq. 8 seems kind of ad hoc, although I acknowledge that there is long-range attraction in the experimental videos. Where does the screening in Eq. 8 come from? What sets the length scale for this? (Perhaps a citation to Ristenpart would be helpful.) Wouldn't Eq. 8 by itself, without particle/particle HI or dipole-dipole interactions, promote crystallization?

(Remarks on code availability)

Reviewer #3

(Remarks to the Author)
The manuscript investigates synchronization phenomena in colloidal Quincke rollers subjected to a DC electric field. By varying confinement geometry, particle density, and field strength, the authors report the emergence of collective oscillatory states and identify distinct synchronization regims. The topic is interesting, as it links active matter physics with non–linear dynamics and synchronization theory.

As experimental platform the authors use Quincke rollers, a well–established active colloid system, and the reported oscillatory states are convincingly documented. The authors employ systematic variation of control parameters and characterize the resulting collective dynamics through frequency distributions, order paramters, and phase diagrams. The work demonstrates the potential of Quincke rollers as a model system for collective oscillator dynamics, which will be of interest to researchers accross soft matter and statistical physics. The manuscript is clearly written and the figures are

informative, enhancing accessibility beyond the immediate field. At the same time, several points require clarification and improvment before a final decision can be made.
1. Although the discussion mentions Kuramoto-type models, the connection remains largely qualitative. A systematic embedding—such as a reduced phase-oscillator model with an explicit coupling function—would significantly enhance the interpretation and help place the results within the broader synchronization literature.
2. Is synchronization bidirectionally coupled to the particle interactions? Or in other words: does the particle denisty depend on the degree of synchronization? In particular in view of generalized swarmalator models, this would be an important point.
3. The novelty of the present findings should be sharpened by explicitly stating how these oscillatory states differ from known roller collectives such as flocks, vortices, or oscillations near electrodes.
4. The phase diagrams provide a useful overview but remain descriptive. Some physical rationale for the observed transition boundaries would add depth—for instance, scaling arguments based on hydrodynamic or electrostatic interaction strengths, or density-dependent coupling ranges.
5. Hydrodynamic couplings are invoked as a possible mechanism, yet they are not quantified. Without controlled pair experiments, simulations, or order-of-magnitude estimates, their role relative to electrostatic or steric effects remains speculative. Even an approximate analysis would strengthen the mechanistic discussion.

In summary, the manuscript reports interesting and carefully executed experiments, but remains primarily descriptive in its current version. The manuscript would largely benefit from a stronger theoretical embedding into existing frameworks and quantification of relevant coupling mechanisms.

(Remarks on code availability)

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)
In the reply, the authors have addressed my questions and concerns, revising the structure of the paper and discussing missing experimental details in the methods section.

After consistent rewriting of the main text, I believe that this manuscript is suitable for publication in Nature Communications in its current form.

(Remarks on code availability)

Reviewer #2

(Remarks to the Author)
In the revised manuscript and response, the authors have satisfactorily addressed the issues I raised. Notably, the authors have substantially expanded their analysis with a reduced-order swarmalator-type model for hydrodynamic interactions between oscillators, providing a mechanistic explanation for their findings (synchronization, alignment, and acceleration) and better connecting their work with the previous literature. The reduced-order model qualitatively reproduces results from the experiments and SD simulations. In addition, the revised SI has a clear explanation of the SD methodology and derivations of the reduced-order model. This will surely help readers who are interested in the technical details.

I approve of publication of the manuscript in its present form.

(Remarks on code availability)
This code was previously published by the Donev group and was featured in various of their papers.

Reviewer #3

(Remarks to the Author)
I am happy to say, that the authors have responded thoroughly and convincingly to all the referee comments, including my own. The revision significantly strengthens the manuscript: the addition of the reduced-order model establishes a clear theoretical framework connecting the results to Kuramoto-type synchronization and XY-like alignment, and the mechanistic role of hydrodynamic interactions is now clearly articulated. Clarifications regarding novelty, finite-size effects, and the relationship to previous Quincke-roller studies further improve the context and impact of the work.
In my assessment, all major concerns have been satisfactorily addressed, and the manuscript now meets the standards for publication in Nature Communications.

(Remarks on code availability)

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In cases where reviewers are anonymous, credit should be given to 'Anonymous Referee' and the source.
The images or other third party material in this Peer Review File are included in the article's Creative Commons license, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons license and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder.
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We gratefully acknowledge the Reviewers for their careful reading of our manuscript and for their constructive recommendations. We have carefully addressed the points that were raised, which we believe has strengthened the manuscript. Our point by point response is provided below in blue text; changes to the manuscript are also highlighted in blue.

Reviewer 1

Through simulations and experiments, the authors study Quincke colloids, which behave as self-oscillating units (active oscillators). In this study, different collective states are observed, including temporal synchronization and directional alignment, termed "synchronematic order". The authors distinguish between fluid-like clusters, where this order is short-range, and crystalline clusters, where the system is characterized by global synchronization and circular alignment.

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I believe that this paper would benefit from a restyling: the authors should introduce sections, distinguishing between introduction, results (perhaps with subsections), and discussion.

Reply: We have added section and subsection headings to guide the reader.

Could the authors clarify what they mean when they write "these spatiotemporal structures arise from different initial conditions ..." or "By varying the initial particle distribution ..."? This point should be explained with more details in the main text and methods. Do these different phases emerge when different packing fractions are considered? The authors should clarify the experimental limitations: are the authors able to control the density in their system? What do the authors mean by the sentence "when initialized within local regions of high density ..."?

Reply: We have expanded the Methods section to discuss the preparation of the initial particle configurations as well as our limited control over these configurations.

The fluid-cluster snapshots (Fig. 1e and Fig. 2a) do not show a cluster with a fluid structure but (apparently) a homogeneous fluid. I suggest that the authors show both an image where the cluster is evident and, additionally, the already present zoom, which focuses on the internal structure (Fig. 2a).

Reply: We have replaced Fig. 1e with a lower magnification image showing the fluid cluster with spatially varying density as well as an inset showing the local particle ordering.

The phase-angle spatial order (exponential decay) resembles the short-range order observed in the velocity field of self-propelled systems (without alignment interactions): L. Caprini, U. Marini Bettolo Marconi, A. Puglisi, "Spontaneous velocity alignment in motility-induced phase separation." *Phys. Rev. Lett.* **124**(7), 078001, 2020. Could the authors comment on these previous results and provide an intuitive argument to explain the origin of the observed phenomenon?

Reply: In the original manuscript, we provide an order-of-magnitude argument to explain the finite correlation length by balancing the strength of hydrodynamic interactions (which promote synchronization and alignment) with that of rotational diffusion. In the revised manuscript, we strengthen these arguments by developing a reduced-order model that describes the evolution of oscillator phases and orientations due to far-field hydrodynamic interactions (see new Section

"Reduced-Order Swarmalator Model for Quincke Oscillators", Figure 5, and Supplementary Notes 6-8). This model reproduces the exponentially decaying synchronematic correlations and enables more direct comparisons to other simplified active matter models, such as that of Caprini et al. (new reference #14). While both systems display exponentially decaying spatial correlations, the physical origins differ fundamentally. Caprini et al. consider self-propelled (polar) active Brownian particles that interact only through excluded-volume repulsion and rotational diffusion, with no alignment interactions. By contrast, our system comprises oscillating (apolar) colloids that behave as nematic rotors with internal phase variables. These rotors interact and align through long-range, hydrodynamic couplings, while rotational diffusion limits the range of synchronematic order. Additionally, non-reciprocal phase–angle interactions couple local oscillation frequencies to density fluctuations, providing a distinct source of disorder absent in polar active-particle systems.

Concerning the synchronematic crystalline phase, is the phenomenon observed a result of finite-size effects? In other words, does this effect disappear for larger cluster sizes? Alternatively, one can wonder if, in the absence of cluster boundaries – say, by considering a homogeneous crystal (in simulations) with periodic boundary conditions – the same synchronization effect appears. I expect that this will not be the case: this point should be clarified. Somehow, this state is reminiscent of the self-reverting vorticity observed in clusters consisting of chiral active particles: L. Caprini, B. Liebchen, H. Löwen, "Self-reverting vortices in chiral active matter." *Commun. Phys.* **7**(1), 153, 2024. The authors may comment on this analogy in the final discussion.

Reply: We have addressed these questions using the new reduced-order model that governs the dynamics of oscillator phases and orientations. Simulations of finite-size crystals show that phase-locked states with circular alignment are stable only below a critical size (Figure 5c,d). The competition between reciprocal phase interactions (which favor synchronization) and non-reciprocal phase interactions (which accelerate oscillations) produces radial phase gradients within the crystals: the phase of oscillators near the center or edge lag behind those in intermediate regions. As the cluster size increases, these gradients grow and eventually destabilize the global phase-locked state.

Simulations on a square lattice with periodic boundary conditions confirm that the circular synchronization observed in finite clusters does not persist in homogeneous systems (see Supplementary Note 7). In these periodic lattices, global phase and orientational order are disrupted by the spontaneous formation of topological defects in both the orientation and phase fields, resembling the defect-mediated disorder observed in the classical XY model. In one-dimensional chains, the same coupling mechanisms give rise to traveling metachronal waves, reminiscent of those seen in models of hydrodynamically coupled cilia.

We have added a comment to the conclusions highlighting connections to similar types of order observed in active matter systems, including that reported by Caprini et al. (new reference #53). Unlike that work, which considers polar, chiral active particles interacting through isotropic potentials, our Quincke oscillators are apolar and achiral, possess an internal phase variable, and interact via anisotropic, non-reciprocal hydrodynamic couplings. These differences in symmetry and interaction type lead to qualitatively distinct forms of collective behavior.

Minor points:

Along with the paper, the authors mention "close-packed oscillating crystals". Actually, I believe that it is more correct that the authors refer to this state as "oscillating clusters with a crystalline structure".

Reply: We have made the requested change.

Typos:
- On page 5, some formulas in line with the text are corrupted, as the formula for r and theta.
- Typo in the Methods: the section title is "Data nalysis" rather than "Data Analysis".
- Several typos are present in Eqs.(2),(3),(4), and (5).
- In formula (8), the minus sign in the argument of the exponential is missing, as well as the minus in the delta Dirac function after Eq.(7).

Reply: We have corrected these errors and revised the Methods section for improved clarity and consistency.

Reviewer 2

The present manuscript is a combined experimental/theoretical study concerning the collective dynamics of oscillatory Quincke rollers. Specifically, the authors study spherical particles that, under an applied electric field, exhibit a back-and-forth rolling motion characterized by frequency, phase, amplitude, and axis **n**. They find that a semi-dilute monolayer of these particles can form liquid-like phases with short-ranged synchronization and clustering (Fig. 1e, Fig. 2b). A more dense monolayer forms rocking crystallites with circular alignment of the particles' rocking directions (Fig. 3b). Numerical modeling reveals that the behavior arises from the interplay of hydrodynamic interactions and dipole-dipole interactions, with an additional role for electroosmotic attraction for the crystallites. Excellent quantitative agreement (Fig. 3d-e) is obtained between the numerics and experiment with seemingly limited fitting.

The study seems thorough and scientifically sound (although I have some questions, as detailed below). Turning to the question of impact/significance: while the actual rocking motion seems, to honest, underwhelming in videos, what I judge the authors to have achieved in this manuscript is linking temporal (synchronization) and spatial (clustering, alignment) forms of ordering. I think this will open new directions in out-of-equilibrium colloidal self-organization. However, to make this case more effectively, the authors need to situate their work with respect to previous work on "swarmalators" and chiral active matter: the manuscript is missing, e.g., citations of D. Levis and J. Lintuvuori. How does the present work go beyond previous work that links flocking and synchronization in circle swimmers? (In addition, I suggest citing the work on AC Quincke rollers from Sano et al.)

Reply: We have added a new section to the paper titled "Reduced-Order Swarmalator Model for Quincke Oscillators" (see also new Figure 5 and Supplemental Notes 6-8). Using the theory of weakly coupled oscillators, we derive and analyze a reduced-order model that describes the coupled evolution of oscillator phases and orientations mediated by far-field hydrodynamic interactions. This model reproduces the main experimental and simulation results and establishes clear links between synchronematic order and classical frameworks for collective dynamics—specifically, Kuramoto-type synchronization, XY-like alignment, and swarmalator models that couple phase and motion. By explicitly incorporating non-reciprocal, anisotropic hydrodynamic couplings, our framework extends previous studies of chiral and polar active systems to the case of apolar, oscillatory colloids.

In light of the Reviewer's suggestions, we have revised both the *Introduction* and *Conclusions* to explicitly connect our work to the studies mentioned. In particular, we have added a citation to

Lintuvuori's work as an example of a collective synchronized state (new reference #3). The revised *Introduction* now compares the features of the reduced-order model derived in this study to both the XY and Kuramoto models, thereby establishing a clearer link to Levis' work (new reference #10). Finally, we have included a new reference to Sano's work on externally driven Quincke rollers under time-periodic forcing (new reference #34).

As a second point: I find the role for hydrodynamic interactions to be very intriguing, but obscured by the presentation of the manuscript. HI are discussed either superficially in the main text (e.g., a cursory mention of vorticity) or buried in formalism/methodology. Overall, a mechanistic understanding of the pairwise interaction of two rollers (interplay of dipole/dipole interactions and HI) is missing. While dipole/dipole interactions are at least well understood, I suggest that the authors could provide at least provide data on the (time-averaged?) flow and vorticity fields sourced by a roller. How do HIs in this instance differ from, e.g., the usual pusher/puller stresslet HIs?

Reply: We have clarified the mechanistic role of hydrodynamic interactions and expanded their presentation in the revised manuscript. The reduced-order model derived using weakly coupled oscillator theory now provides explicit interaction functions averaged over the oscillation limit cycles, yielding a simplified yet quantitative description of how hydrodynamic coupling drives synchronization and alignment among oscillators (new Section "Reduced-Order Swarmalator Model for Quincke Oscillators"). This framework captures the emergent collective behavior observed in experiments and full Stokesian Dynamics simulations while elucidating the fundamental contribution of hydrodynamics.

In contrast to classical pusher–puller stresslet models, the hydrodynamic disturbances created by Quincke oscillators are time-dependent and strongly influenced by the nearby wall. While rotlets and stresslets in unbounded fluids decay as r^{-2}, the leading order disturbance of a rotating and translating sphere above a plane wall scales as r^{-3}. When averaged over the oscillation cycle, however, these leading-order contributions can partially or fully cancel—yielding effective hydrodynamic interactions among oscillator positions that decay even faster, as r^{-4}. This distinction explains why long-range coupling in phase and orientation dominates the collective behavior, whereas positional interactions play only a secondary role.

More detailed comments/questions:

Section headings might help with navigating this manuscript.

Reply: We have added section and subsection headings to guide the reader.

Please elaborate on the point on p. 5 about the value of the covariance suggesting a common origin of synchronization and alignment.

Reply: The shape of the central peak in Fig. 2c reflects the relative strength of synchronization and alignment between neighboring oscillators. If these two forms of order arose from different mechanisms—such as hydrodynamic versus electrostatic interactions—their relative strengths would likely differ, producing an anisotropic peak elongated along one axis. The observed near-isotropic peak implies that synchronization and alignment occur with comparable magnitude and likely share a common origin. Our reduced-order swarmalator model supports this interpretation, showing that hydrodynamic interactions drive both synchronization and alignment.

Please clarify the argument on p. 9 concerning why the finite amplitude of oscillations limits the diameter for rigid body motion. (Where does the factor of π come from?)

Reply: The oscillatory displacement of a particle has a peak-to-peak amplitude A (Fig. 1c), which varies with the particle's local environment—smaller near the cluster center and largest for an isolated oscillator. Because a particle travels a distance A over half of an oscillation period (π/ω), its mean speed is $A\omega/\pi$. This speed reaches its maximum value $A_{\max}\omega/\pi$ for a free oscillator with amplitude $A_{\max}$, corresponding approximately to the distance a particle moves as its electric dipole relaxes from an antiparallel to a parallel orientation in the field (see reference #31). This finite amplitude sets the upper limit on the tangential velocity that individual oscillators can sustain and therefore constrains the maximum radius over which rigid-body-like rotation can persist in synchronematic crystals.

Concerning the role for q in the numerics: is it correct that $q \neq 0$ in Eqs. 2-4, since there is represents surface charge, but is assumed to be zero for particle/particle interaction, due to screening by ions?

Reply: Indeed, while the oscillator model predicts that particles acquire a net charge q, this charge would generate strong Coulomb repulsion between neighboring particles. Experimentally, no such repulsion is observed, suggesting that the particle charge is screened by the surrounding electrolyte. While excluded in the oscillator model, free charge in the electrolyte is likely responsible for such screening and for the electroosmotic flows hypothesized to drive cluster formation.

The introduction of Eq. 8 seems kind of ad hoc, although I acknowledge that there is long-range attraction in the experimental videos. Where does the screening in Eq. 8 come from? What sets the length scale for this? (Perhaps a citation to Ristenpart would be helpful.) Wouldn't Eq. 8 by itself, without particle/particle HI or dipole-dipole interactions, promote crystallization?

Reply: We have added a citation to Ristenpart et al. (new reference #43) describing how electroosmotic and electrohydrodynamic flows near conducting electrodes can produce long-range attractions between polarized particles. The screened form of Eq. 8 is indeed phenomenological. While this term alone can promote crystallization, its inclusion in the simulations allows us to investigate how such confinement couples with hydrodynamic and electrostatic interactions to shape the observed synchronematic order—namely, phase synchronization and circular alignment. These behaviors do not arise from the confining potential alone but emerge from its interplay with the oscillatory motion and hydrodynamic coupling among the particles.

Reviewer 3

The manuscript investigates synchronization phenomena in colloidal Quincke rollers subjected to a DC electric field. By varying confinement geometry, particle density, and field strength, the authors report the emergence of collective oscillatory states and identify distinct synchronization regims. The topic is interesting, as it links active matter physics with non–linear dynamics and synchronization theory.

As experimental platform the authors use Quincke rollers, a well–established active colloid system, and the reported oscillatory states are convincingly documented. The authors employ systematic variation of control parameters and characterize the resulting collective dynamics through frequency distributions, order paramters, and phase diagrams. The work demonstrates the potential of Quincke rollers as a model system for collective oscillator dynamics, which will be of interest to researchers accross soft matter and statistical physics. The manuscript is clearly written and the figures are informative, enhancing accessibility beyond the immediate field. At the same time, several points require clarification and improvment before a final decision can be made.

1. Although the discussion mentions Kuramoto-type models, the connection remains largely qualitative. A systematic embedding—such as a reduced phase-oscillator model with an explicit coupling function—would significantly enhance the interpretation and help place the results within the broader synchronization literature.

Reply: We have added a new section to the paper titled "Reduced-Order Swarmalator Model for Quincke Oscillators" (see also new Figure 5 and Supplemental Notes 6-8). Using the theory of weakly coupled oscillators, we derive and investigate a reduced-order model that describes the coupled evolution of oscillator phases and orientations due to far field hydrodynamic interactions. The reduced model reproduces the main experimental and simulation results and clarifies the connection to Kuramoto-type synchronization and XY-like alignment models, while highlighting the unique role of non-reciprocal interactions in our system.

2. Is synchronization bidirectionally coupled to the particle interactions? Or in other words: does the particle denisty depend on the degree of synchronization? In particular in view of generalized swarmalator models, this would be an important point.

Reply: In the reduced-order model based on far-field hydrodynamic and electrostatic interactions, there is bidirectional coupling between the oscillator phases and orientations—between synchronization and alignment. The strength of these couplings depends on the oscillator positions, since pairwise phase-orientation interactions decay as d^{-3} with separation d. By contrast, interactions among oscillator positions are subdominant, scaling as d^{-4}, and are not explicitly included in our model. Thus, to leading order, the particle density does not depend on the degree of synchronization or alignment; however, such couplings are likely present at higher order.

3. The novelty of the present findings should be sharpened by explicitly stating how these oscillatory states differ from known roller collectives such as flocks, vortices, or oscillations near electrodes.

Reply: In the revised manuscript, we have expanded the Conclusions to clarify how the oscillatory states reported here differ from previously described Quincke roller collectives such as flocks, vortices, and oscillations near electrodes. Specifically, we emphasize that synchronematic colloids exhibit a distinct form of activity-induced order arising from hydrodynamic coupling between self-oscillating, apolar, and achiral units, whose collective dynamics are highly sensitive to geometric organization.

4. The phase diagrams provide a useful overview but remain descriptive. Some physical rationale for the observed transition boundaries would add depth—for instance, scaling arguments based on hydrodynamic or electrostatic interaction strengths, or density-dependent coupling ranges.

Reply: The manuscript distinguishes qualitatively between dilute, fluid, and crystalline synchronematic states rather than presenting a quantitative phase diagram. The observed transitions reflect a balance between long-range hydrodynamic coupling, which promotes synchronization and alignment, and short-range electroosmotic attraction, which drives local densification. While our current Stokesian Dynamics and reduced-order swarmalator models capture the mechanisms underlying synchronematic order within each regime, they do not resolve the processes that determine when or where crystallization occurs.

5. Hydrodynamic couplings are invoked as a possible mechanism, yet they are not quantified. Without controlled pair experiments, simulations, or order-of-magnitude estimates, their role relative to electrostatic or steric effects remains speculative. Even an approximate analysis would strengthen the mechanistic discussion.

Reply: In the original manuscript, Stokesian Dynamics simulations performed without electrostatic interactions accurately reproduced the experimental results, providing strong (albeit indirect) evidence that hydrodynamic interactions underlie synchronematic ordering. In the revised manuscript, the new reduced-order model makes this connection explicit by deriving the phase and orientation couplings from far-field hydrodynamic interactions.

In summary, the manuscript reports interesting and carefully executed experiments, but remains primarily descriptive in its current version. The manuscript would largely benefit from a stronger theoretical embedding into existing frameworks and quantification of relevant coupling mechanisms.

Reply: We thank the Reviewer for their constructive feedback, which prompted substantial revisions to strengthen the theoretical framework and clarify the physical mechanisms underlying synchronematic order.